
FIDELITY IS NOT UTILITY: A CONTROLLED DISSOCIATION IN LEARNED IMAGE TRANSLATION

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ABSTRACT

Image translation for harmonization and domain adaptation is routinely selected on distributional fidelity metrics such as the Fréchet Inception Distance (FID), on the assumption that a more target-like image is more useful: a team can lower FID and ship a model that is worse at the task. That fidelity can mis-rank generators is known; we make a sharper, mechanistic claim: the utility loss is a property of the learned distribution-matching *objective* itself, not of any architecture, which is why it survives the move from GANs to diffusion. Our instrument is a *frozen* downstream evaluator scored alongside FID on identical inputs, plus a matched non-learned control. On GTA5→Cityscapes ($N=1000$, frozen SegFormer), CycleGAN more than halves FID (137.2→65.5) while mIoU falls 0.380→0.167; the non-learned control preserves mIoU (0.363) without improving FID. Sweeping SDEdit’s noise strength traces a monotonic frontier: FID 145.1→88.0 as mIoU falls 0.323→0.106, with seed standard deviation at most 0.005 mIoU. Two controls close the obvious escapes: with an empty prompt, fidelity is worse than the untranslated source (FID 154.2) yet mIoU still collapses (0.170), decoupling the damage from any fidelity gain; and edge-conditioned ControlNet does not escape the frontier (mIoU 0.221 at FID 133.2, versus 0.289 at matched FID). Multi-site brain-tumor MRI reproduces it: frozen nnU-Net tumor Dice falls 0.682→0.064 under learned harmonization but holds at 0.669 non-learned. Across both generative families and both domains, the low-FID/high-utility corner stayed empty. We do not claim fidelity is uninformative, only that it is non-monotone in utility once a learned prior enters the loop; our contribution is a measurement and a protocol, not a method.

1 Introduction

Distribution-matching translation is trained and evaluated to make outputs *look like* a target domain. The implicit promise is that looking more like the target makes outputs more *useful*—for example, that harmonizing site B to site A helps a model trained on A . We test that promise directly and find it false for learned translation.

Our contributions are:

- A **controlled protocol** for separating fidelity from utility: a frozen downstream evaluator plus a matched non-learned control, measuring FID and the task metric on identical inputs (Section 3).
- Evidence that the fidelity–utility dissociation is a property of the **learned distribution-matching objective**, not a single architecture—it reproduces under both a GAN (CycleGAN) and a diffusion model (SDEdit).
- A **continuous characterization** via SDEdit strength: a monotonic FID-vs-mIoU frontier with tight seed variance, an empty-prompt ablation that decouples utility loss from fidelity gain, and a structure-preserving (ControlNet) baseline that fails to escape the frontier.
- **Cross-domain** generality: the same dissociation in driving scenes and in clinical brain-tumor MRI.

2 Related Work

Three separately-established observations converge in our claim; we unify them.

Fidelity metrics mis-rank generators for downstream tasks. Konz et al. [2024] show no perceptual metric reliably tracks segmentation quality in medical translation; Dutt et al. [2025] show low FID does not predict clinical utility; and Kynkäänniemi et al. [2023] show FID is gameable by aligning ImageNet-class statistics. These are *metric-evaluation* studies; we add a learned/non-learned control and a second generative family and domain.

Distribution-matching translation alters task content. Cohen et al. [2018] show CycleGAN-style matching can hallucinate or remove tumors to fit the target distribution, and semantic-consistency losses [Hoffman et al., 2018, Peng et al., 2023] exist precisely to patch this. We show that *without* such non-distributional anchoring the pure learned objective degrades utility, reframing those losses as evidence for our thesis.

Non-learned alignment rivals learned translation. Yang and Soatto [2020] reach strong segmentation adaptation with a non-learned spectral swap. We use histogram/color transfer as the analogous control and, unlike FDA, measure the fidelity axis explicitly to expose the dissociation. Richter et al. [2022] likewise treat a segmentation-stability metric as a guardrail that realism must not break.

Positioning. No prior work runs the clean 2×2 {learned, non-learned} $\times$ {fidelity $\downarrow$?, utility $\downarrow$?} as its thesis, ties the failure to the objective across both a GAN *and* a diffusion model, and does so across driving and medicine. This is a measurement/positioning contribution, not a new architecture.

3 Setup: Separating Fidelity from Utility

Given source images x and a translation T , we evaluate $T(x)$ on two axes. **Fidelity** is $\text{FID}(T(x), \text{target})$ via clean-fid [Parmar et al., 2022]; lower is closer to the target. **Utility** is a *frozen* downstream model f (never retrained) applied to $T(x)$ and scored against ground truth; because f is fixed, any change in its score is attributable purely to T . We compare four families of T : (i) identity (raw source, reference); (ii) non-learned (histogram / color transfer); (iii) learned-GAN (CycleGAN, Zhu et al., 2017); and (iv) learned-diffusion (SDEdit, Meng et al., 2022, over a sweep of noise strengths, plus ControlNet-Canny, Zhang et al., 2023). FID is compared only at equal sample size, which it is known to bias.

4 Experiments

4.1 Driving: GTA5 $\rightarrow$ Cityscapes

The source is GTA5 [Richter et al., 2016] and the target is Cityscapes [Cordts et al., 2016]. The frozen evaluator is a Cityscapes-finetuned SegFormer-b4 [Xie et al., 2021] (19 classes); utility is mIoU. We use $N=1000$ GTA5 images per condition, with the Cityscapes validation set (500 images) as the fixed FID reference, and identical evaluator and clean-fid (GPU) for all conditions. SDEdit runs on Stable Diffusion [Rombach et al., 2022] and rows report mean $\pm$ std over three seeds {42, 43, 44}. Results are in Table 1 and Figure 1.

Findings. Figure 2 shows representative translations. (a) Learned translation lowers FID relative to raw while collapsing mIoU by 15–72%; CycleGAN more than halves FID (137 $\rightarrow$ 65) at mIoU 0.167. (b) The non-learned baseline does not improve FID yet preserves mIoU (−1.8%). (c) SDEdit strength traces a **monotonic frontier** with tight seed variance ($\sigma(\text{mIoU}) \leq 0.005$): as strength rises, FID falls (145 $\rightarrow$ 88) and mIoU falls with it (0.323 $\rightarrow$ 0.106); no learned setting reaches the ideal corner, occupied only by raw/non-learned. (d) **Empty-prompt ablation.** With no text prompt, FID is *worse* than raw (154.2 $>$ 137.2) yet mIoU still collapses (0.170)—the utility loss is the diffusion prior itself, not text guidance, and occurs even absent any fidelity gain. (e) **Structure-preserving baseline.** Conditioning on the source image’s Canny edges (ControlNet-Canny) does *not* rescue utility: it lands on the learned frontier (FID 133.2, mIoU 0.221 $\pm$ 0.002), Pareto-dominated by SDEdit at equal FID (SDEdit 0.30: FID 133.1, mIoU 0.289). Edge structure is not the task-relevant signal; only conditioning on the segmentation *map* [Peng et al., 2023] recovers utility, and only by injecting the labels—the non-distributional constraint our thesis requires. A local $N=200$ run reproduces the same ordering on an independent machine.

Table 1: GTA5→Cityscapes, $N=1000$, frozen SegFormer-b4 + clean-fid. SDEdit rows are mean $\pm$ std over three seeds. Learned translation lowers FID while collapsing mIoU; the non-learned baseline preserves mIoU without improving FID; no learned setting reaches the low-FID/high-mIoU corner.

condition	family	FID $\downarrow$	mIoU $\uparrow$	Δ mIoU
raw GTA5	baseline	137.2	0.380	—
color match	non-learned	165.9	0.363	−0.018
SDEdit 0.20	diffusion	145.1 $\pm$ 1.8	0.323 $\pm$ 0.003	−0.057
SDEdit 0.30	diffusion	133.1 $\pm$ 2.1	0.289 $\pm$ 0.002	−0.092
SDEdit 0.40	diffusion	116.4 $\pm$ 1.1	0.237 $\pm$ 0.003	−0.143
SDEdit 0.50	diffusion	100.3 $\pm$ 1.4	0.194 $\pm$ 0.005	−0.187
SDEdit 0.60	diffusion	90.8 $\pm$ 2.0	0.150 $\pm$ 0.001	−0.231
SDEdit 0.70	diffusion	88.0 $\pm$ 0.4	0.106 $\pm$ 0.001	−0.274
CycleGAN	learned-GAN	65.5	0.167	−0.214
SDEdit 0.50, empty prompt	diffusion (abl.)	154.2 $\pm$ 1.9	0.170 $\pm$ 0.003	−0.211
ControlNet-Canny	diffusion + edge	133.2 $\pm$ 0.3	0.221 $\pm$ 0.002	−0.159

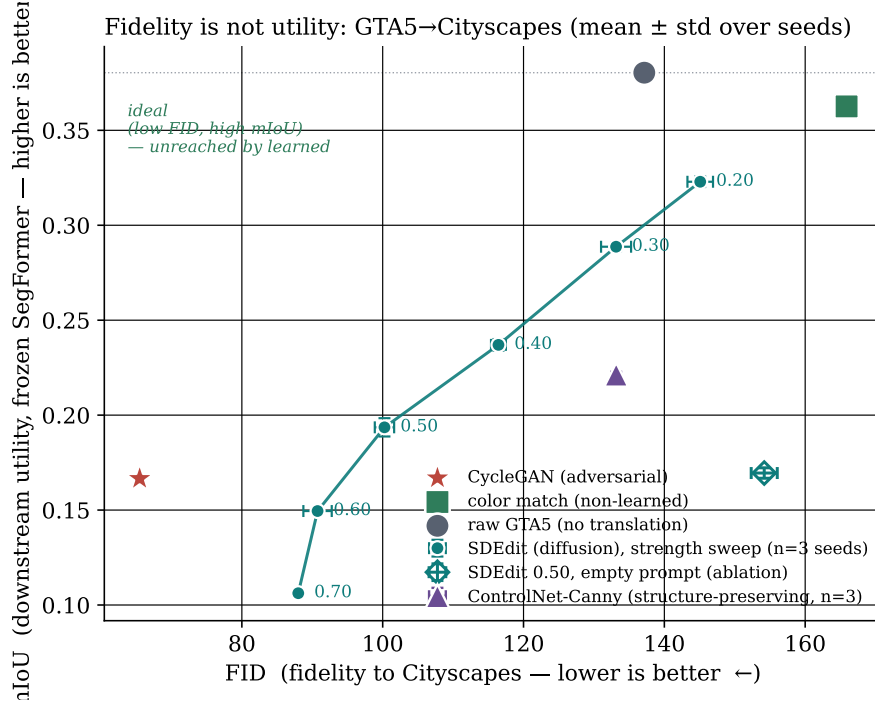


Figure 1: The fidelity–utility plane for GTA5→Cityscapes (mean $\pm$ std over seeds). The SDEdit strength sweep traces a monotonic frontier: as strength rises, FID falls and mIoU falls with it. Every learned point (CycleGAN, all SDEdit strengths, the empty-prompt ablation, and edge-conditioned ControlNet) sits well below raw and the non-learned baseline on utility; the ideal low-FID/high-mIoU corner is empty.

4.2 Medical: Multi-Site Brain-Tumor MRI

The frozen evaluator is an nnU-Net [Isensee et al., 2021] trained on BraTS [Menze et al., 2015] and frozen; utility is tumor Dice on the external UPenn-GBM site [Bakas et al., 2022] ($n=103$). The learned harmonizer is SA-CycleGAN-2.5D, a self-attention CycleGAN over 2.5D multi-modal MRI. Table 2 reports the result.

Findings. Learned harmonization collapses external-site Dice (0.682→0.064) while the non-learned baseline preserves it (0.669). Content/segmentation-consistency losses reduce but do not eliminate the loss (0.205 $\ll$ 0.682), consistent with the driving result that the degradation is intrinsic to the learned objective. A second external site (RHUH-GBM, Cepeda et al., 2023, $n=40$) confirms the direction (0.718→0.599, −16.6%). We note two evaluation

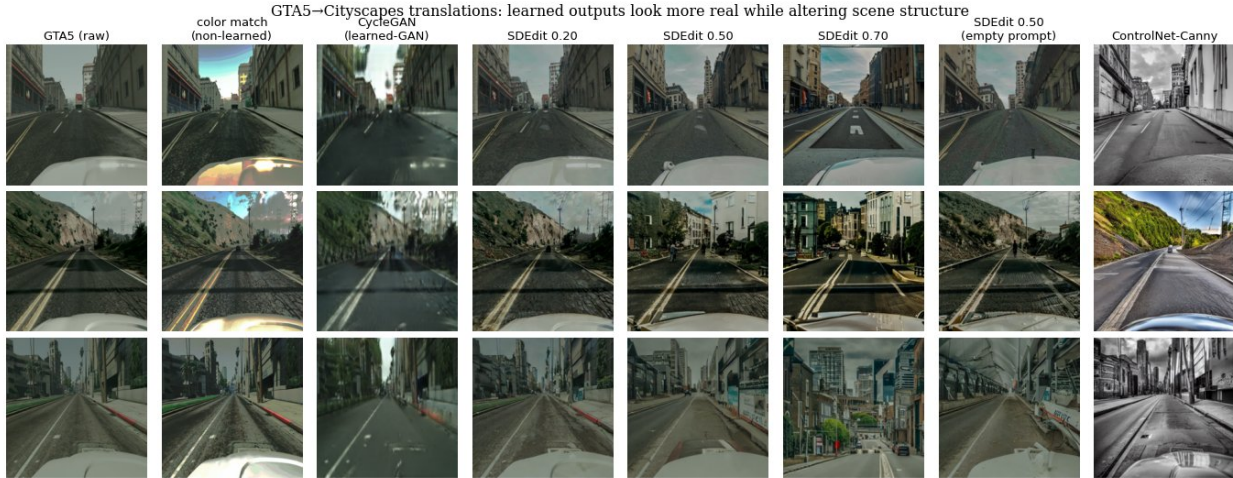


Figure 2: Qualitative GTA5→Cityscapes translations (three scenes, seed 42). Left to right: raw source, non-learned color match, CycleGAN, SDEdit at increasing strength, the empty-prompt SDEdit, and edge-conditioned ControlNet. As the learned translations become more Cityscapes-like they visibly hallucinate and alter scene structure—buildings, lane markings, added objects—that the frozen segmenter depends on, while the non-learned baseline leaves structure intact.

Table 2: External UPenn-GBM tumor Dice under a frozen BraTS-trained nnU-Net. Learned harmonization collapses Dice even though it makes images more source-like; the non-learned baseline preserves it.

input to frozen nnU-Net	family	Dice (fg. mean)
BraTS test (in-domain ref.)	—	0.771
UPenn raw (external)	baseline	0.682
UPenn + histogram match	non-learned	0.669
UPenn + learned harmonization	learned-GAN	0.064
UPenn + symmetric seg-consistency	learned-GAN (+ content)	0.205

protocols exist: the headline numbers use the frozen independent nnU-Net; a separate method-development evaluator ranks the symmetric loss as the best-preserving *learned* harmonizer, which belongs to a companion method paper.

5 Analysis

The learned objective optimizes appearance under a distribution/adversarial or denoising target with no term guaranteeing task-relevant signal survives. SDEdit strength is precisely the amount of learned prior injected, and utility falls monotonically with it. The empty-prompt result decouples utility loss from fidelity gain: passing through the prior damages task content even when the output is *less* target-like. Non-learned transforms, lacking a learned prior, cannot introduce this failure; edge-conditioned ControlNet, injecting structure but not semantics, does not avoid it.

6 Limitations

FID is sample-size biased; all conditions use the same N (1000 translated vs the 500-image Cityscapes validation set as the fixed reference), and SDEdit points report mean $\pm$ std over three seeds ($\sigma(\text{mIoU}) \leq 0.005$, $\sigma(\text{FID}) \leq 2.1$), so the ordering is robust to both sample size and seed. The structure-preserving baseline uses edge (Canny) conditioning; a segmentation-map-conditioned variant would recover utility only by injecting the labels, which concedes the thesis rather than refuting it.

7 Conclusion

Fidelity is not utility. Rewarding distributional realism can actively harm the downstream task for learned translation, across architectures and domains. We recommend reporting a frozen-evaluator utility axis alongside any fidelity metric, and treating non-learned baselines as first-class controls.

Reproducibility Statement

All conditions regenerate from a single command; seeds are fixed and logged; frozen evaluators are referenced by checkpoint id (SegFormer-b4, nnU-Net); FID uses clean-fid at a fixed N . Code, configs, and the $N=1000$ result set are released with the paper.

References

- Spyridon Bakas et al. The university of pennsylvania glioblastoma (UPenn-GBM) cohort: Advanced MRI, clinical, genomics, and radiomics. *Scientific Data*, 9(453), 2022.
- Santiago Cepeda et al. RHUH-GBM: A radiomics and clinical dataset of glioblastoma from río hortega university hospital. *Data in Brief*, 2023.
- Joseph Paul Cohen, Margaux Luck, and Sina Honari. Distribution matching losses can hallucinate features in medical image translation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2018.
- Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. The Cityscapes dataset for semantic urban scene understanding. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- Raman Dutt et al. CheXGenBench: A unified benchmark for fidelity, privacy and utility of synthetic chest radiographs. *arXiv preprint arXiv:2505.10496*, 2025.
- Judy Hoffman, Eric Tzeng, Taesung Park, Jun-Yan Zhu, Phillip Isola, Kate Saenko, Alexei Efros, and Trevor Darrell. CyCADA: Cycle-consistent adversarial domain adaptation. In *International Conference on Machine Learning (ICML)*, 2018.
- Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021.
- Nicholas Konz, Yuwen Chen, Hanxue Gu, Haoyu Dong, and Maciej A. Mazurowski. Rethinking perceptual metrics for medical image translation. *arXiv preprint arXiv:2404.07318*, 2024.
- Tuomas Kynkäänniemi, Tero Karras, Miika Aittala, Timo Aila, and Jaakko Lehtinen. The role of ImageNet classes in Fréchet inception distance. In *International Conference on Learning Representations (ICLR)*, 2023.
- Chenlin Meng, Yutong He, Yang Song, Jiaming Song, Jiajun Wu, Jun-Yan Zhu, and Stefano Ermon. SDEdit: Guided image synthesis and editing with stochastic differential equations. In *International Conference on Learning Representations (ICLR)*, 2022.
- Bjoern H. Menze et al. The multimodal brain tumor image segmentation benchmark (BRATS). *IEEE Transactions on Medical Imaging*, 34(10):1993–2024, 2015.
- Gaurav Parmar, Richard Zhang, and Jun-Yan Zhu. On aliased resizing and surprising subtleties in GAN evaluation. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- Duo Peng, Ping Hu, Qihong Ke, and Jun Liu. Diffusion-based image translation with label guidance for domain adaptive semantic segmentation. In *IEEE International Conference on Computer Vision (ICCV)*, 2023.
- Stephan R. Richter, Vibhav Vineet, Stefan Roth, and Vladlen Koltun. Playing for data: Ground truth from computer games. In *European Conference on Computer Vision (ECCV)*, 2016.
- Stephan R. Richter, Hassan Abu AlHaija, and Vladlen Koltun. Enhancing photorealism enhancement. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2022.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

- Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M. Alvarez, and Ping Luo. SegFormer: Simple and efficient design for semantic segmentation with transformers. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- Yanchao Yang and Stefano Soatto. FDA: Fourier domain adaptation for semantic segmentation. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In *IEEE International Conference on Computer Vision (ICCV)*, 2023.
- Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A. Efros. Unpaired image-to-image translation using cycle-consistent adversarial networks. In *IEEE International Conference on Computer Vision (ICCV)*, 2017.